

MATH 3115 MSc & SED COMPLEX ANALYSIS

Lecture 7: The Cauchy Integral Formula

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19th Februry, 2025

Lesson Outcome

By the end of this lesson you should be able to:

- Define Integral Triangle Inequality .
- Apply Integral Triangle Inequality in evaluating contour integrals.
- Describe Simply connected Domain
- Describe multiply connected Domain
- Solve problems involving cauchy Goursat theorem

Cauchy Integral Formula

- This theorem states that the value of an analytic function f can be represented by a certain contour integral.
- The n th derivative $f^{(n)}(z)$ will have a similar representation.
- Cauchy Integral Formulae will also be convenient tools for evaluating certain contour integrals

Cauchy Integral formula CONT'D

Theorem 1

Let f be analytic in the simply connected domain D , and let C be a simple closed positively oriented contour that lies in D if z_0 is a point that lies interior to C , then

$$f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0} dz$$

Example 1

Show that $\int_C \frac{e^z}{z-1} dz = 2\pi ei$ where c is the circle $|z| = 2$ with positive orientation

Solution

- Identifying the Singularities: From Integrand $\frac{e^z}{z-1}$ $z_0 = 1$ since denominator $z_1 - 1 = 0$ at $z_0 = 1$
- The contour C is the circle $|z| = 2$, implies it consists of all points z such that $|z| \leq 2$.
- The point $z = 1$ clearly lies inside this contour because $|1| = 1$, and $1 < 2$.
- So the singularity $z = 1$ is inside the contour.

Solution CONT'D

By Cauchy's Integral Formula, the integrand $\frac{e^z}{z-1}$ has a simple pole at $z = 1$, hence

$$f(z_0) = \frac{1}{2\pi i} \int_C \frac{e^z}{z-1} dz$$

But $f(z) = e^z = e^1$ since $z_0 = 1$ lies in the interior to C .
So implies that

$$e^1 = \frac{1}{2\pi i} \int_C \frac{e^z}{z-1} dz$$

Multiply both sides $2\pi i$

$$\int_C \frac{e^z}{z-1} dz = 2\pi i e$$

Example 2

Show that $\int_C \frac{\sin z}{4z + \pi} dz = \frac{-\sqrt{2}}{4} \pi i$ where C is the circle $|z| = 1$ with positive orientation

Solution

- From Integrand $\frac{\sin z}{4z + \pi}$, the denominator suggest the pole where $4z + \pi = 0$, or equivalently, at $z = -\frac{\pi}{4}$.
- This is a simple pole at $z_0 = -\frac{\pi}{4}$.
- By Cauchy's Integral Formula, we need the integrand to be of the form $\frac{f(z)}{z - z_0}$
- Since $f(z)$ is an analytic function, and z_0 is the location of the singularity (the pole).
- Then we rewrite the denominator $4z + \pi$ as $4\left(z + \frac{\pi}{4}\right)$:

$$\frac{\sin z}{4z + \pi} = \frac{\sin z}{4\left(z + \frac{\pi}{4}\right)}$$

Solution

- So we have

$$\int_C \frac{\sin z}{4z + \pi} dz = \frac{1}{4} \int_C \frac{\sin z}{z + \frac{\pi}{4}} dz$$

- By Cauchy's Integral Formula:

$$f(z_0) = \sin\left(-\frac{\pi}{4}\right) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0} dz$$

- $\int_C \frac{\sin z}{z + \frac{\pi}{4}} dz = 2\pi i \cdot \sin\left(-\frac{\pi}{4}\right)$. But $\sin\left(-\frac{\pi}{4}\right) = -\frac{\sqrt{2}}{2}$

- Then

$$\int_C \frac{\sin z}{z + \frac{\pi}{4}} dz = 2\pi i \cdot -\frac{\sqrt{2}}{2}$$

Solution

- From original integral

$$\int_C \frac{\sin z}{4z + \pi} dz = \frac{1}{4} \int_C \frac{\sin z}{z + \frac{\pi}{4}} dz = \frac{1}{4} \cdot (-\sqrt{2}\pi i)$$

$$\int_C \frac{\sin z}{4z + \pi} dz = \frac{-\sqrt{2}\pi i}{4}$$

Example

Find the value of the integral $\int_C \frac{(9z^2 - 1z + 4)}{z(z^2 + 1)} dz$ where C is the circle $|z| = 2$ and the integral along C is taken in the positive direction.

Solution

- By using partial fraction decomposition, we have $\frac{9z^2 - iz + 4}{z(z^2 + 1)} = \frac{A}{z} + \frac{B}{z+i} + \frac{C}{z-i}$ where A , B , and C are constants
- The contour encircles the poles at $z=0$, $z=i$, and $z=-i$.
- Decomposing the denominator

$$z(z^2 + 1) = z(z - i)(z + i)$$

$$\frac{9z^2 - z + 4}{z(z^2 + 1)} = \frac{A}{z} + \frac{B}{z - i} + \frac{C}{z + i}$$

Solution

- Eliminate the denominator by multiply both sides of the equation by $z(z^2 + 1) = z(z - i)(z + i)$
- So we have

$$9z^2 - z + 4 = A(z - i)(z + i) + Bz(z + i) + Cz(z - i)$$

- Expanding both sides

$$A(z - i)(z + i) = A(z^2 + 1)$$

$$Bz(z + i) = Bz^2 + Bzi$$

$$Cz(z - i) = Cz^2 - Czi$$

$$9z^2 - z + 4 = A(z^2 + 1) + Bz^2 + Bzi + Cz^2 - Czi$$

Solution

- Equating coefficient

$$z^2 : 9 = A + B + C$$

$$z : -1 = B - C$$

$$4 = A$$

- Substitute $A = 4$ into the other two equations: 1. $9 = 4 + B + C$ implies $B + C = 5$ 2. $-1 = B - C$

We now have the system:

$$B + C = 5$$

$$B - C = -1$$

$$(B + C) + (B - C) = 5 + (-1)$$

$$2B = 4 \Rightarrow B = 2$$

Now substitute $B = 2$ into $B + C = 5$:

$$2 + C = 5 \Rightarrow C = 3$$

- We have $A = 4$, $B = 2$, and $C = 3$.
- So the partial fraction decomposition is:

$$\frac{9z^2 - z + 4}{z(z^2 + 1)} = \frac{4}{z} + \frac{2}{z - i} + \frac{3}{z + i}$$

- The integral becomes

$$\int_C \frac{9z^2 - z + 4}{z(z^2 + 1)} dz = \int_C \left(\frac{4}{z} + \frac{2}{z - i} + \frac{3}{z + i} \right) dz$$

- Since the contour C is the circle $|z| = 2$, each of these terms has a simple pole inside the contour.
- $\int_C \frac{4}{z}$ around C is $2\pi i \times 4 = 8\pi i$.
- $\int_C \frac{2}{z-i}$ around C is $2\pi i \times 2 = 4\pi i$.
- $\int_C \frac{3}{z+i}$ around C is $2\pi i \times 3 = 6\pi i$.
- $8\pi i + 4\pi i + 6\pi i = 18\pi i$

THE END